

# Project meeting 2: Synthesising academic literature

Minidiss 26/27 with Dr. Elizabeth Pankratz  
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## 1 Extracting key info from academic writing

- Focus on the information that's relevant/important to you. It's OK to ignore details that aren't relevant.
- The challenge: how do you know what information is relevant/important?

### 1.1 Read with a specific target question in mind

- Different sections of your literature review might focus on different questions, e.g.:
  - How does statistical learning work?
  - How is autistic cognition different from allistic cognition?
  - Do autistic people perform better or worse than allistic people on statistical learning tasks? ← today's question

### 1.2 Know/look up the key terminology

**Transitional probability (TP):** the probability that, once you've heard one syllable, another specific syllable will follow

**Distributional cues:** how often units occur, alone or with specific other units, in your input (TPs are one type of distributional cue)

**Statistical learning:** the process of using distributional cues like TPs to recognise patterns in your input

**Word segmentation:** the process of using distributional cues like TPs to identify boundaries between words (sometimes used synonymously with "statistical learning")

**Implicit learning:** learning without explicit instruction (e.g., by passively listening; statistical learning is one type of implicit learning)

### 1.3 For each paper, use this three-step strategy

(hat tip to Dr. Jolène van der Mescht for this method)

#### 1. What did the authors do?

- What was their question?
- What methods did they use? (experiment, survey, meta-analysis, theoretical argumentation, etc.)

#### 2. What did they find?

- What were their results?
- What did they conclude from the results?
- If they present lots of results, prioritise the relevant ones for your target question.

#### 3. Your critical comment / evaluation / "so what?", based on your target question

- Do you think their conclusions are reasonable, based on what they found?
- What implications do you think their findings have?
- What predictions would you make for your study, based on their findings?
- What questions are not answered by this paper?

## 1.4 Example abstract (abbreviated) from Bell et al. (2025)

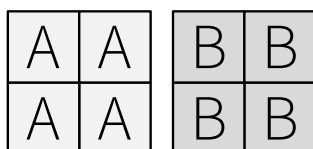
Statistical learning, the ability to detect and extract statistical regularities from the environment, has been proposed as a key mechanism underlying language, social, and cognitive development. Numerous studies have examined statistical learning abilities in autistic individuals to test the hypothesis that differences contribute to the behavioral presentation of autism spectrum disorder (ASD). Findings have been inconsistent, with variations in methodology, sensory modality, and participant characteristics complicating the interpretation of results. The current study presents a systematic review of statistical, implicit, and procedural learning studies in autism, considering how statistical learning abilities vary across (a) modality (e.g., auditory versus visual), (b) methodology (e.g., behavioral versus neuroimaging), and (c) task design, and considering the influence of language and cognitive abilities. Results across 37 studies in visual and auditory modalities indicate few behavioral differences in statistical learning abilities (with the exception of slowed reaction times in autism), and that learning may benefit from extended exposure and explicit cues. In contrast, neuroimaging findings reveal substantial variability in the neural mechanisms implicated in these tasks, with evidence suggesting compensatory cognitive processing in some autistic samples. Individual differences in language, cognitive abilities, and autism-related traits strongly influence statistical learning outcomes. [...]

| What did they do? | What did they find? | Critical evaluation |
|-------------------|---------------------|---------------------|
|                   |                     |                     |

## 2 Activity: Synthesising abstracts

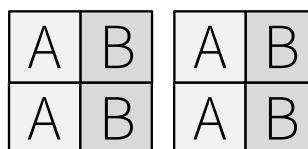
### Phase 1: Expert groups

Work together to understand one abstract.



### Phase 2: Jigsaw groups

Synthesise abstracts:  
How do they relate?



- When reasoning about how papers relate, start by asking questions like:
  - Do they address the same/similar questions?
  - Do they use the same/similar methods?
  - Do they arrive at the same/similar conclusions?

## References and recommended reading

- Bell, R. R., Thomas, H. R., Saffran, J. R., & Eigsti, I.-M. (2025). A systematic review of statistical learning in autism spectrum disorder. *Molecular Autism*, 17(1), 2. <https://doi.org/10.1186/s13229-025-00697-7>
- Haebig, E., Saffran, J. R., & Ellis Weismer, S. (2017). Statistical word learning in children with autism spectrum disorder and specific language impairment. *Journal of Child Psychology and Psychiatry*, 58(11), 1251–1263. <https://doi.org/10.1111/jcpp.12734>
- Obeid, R., Brooks, P. J., Powers, K. L., Gillespie-Lynch, K., & Lum, J. A. G. (2016). Statistical Learning in Specific Language Impairment and Autism Spectrum Disorder: A Meta-Analysis. *Frontiers in Psychology*, 7, 1–18. <https://doi.org/10.3389/fpsyg.2016.01245>